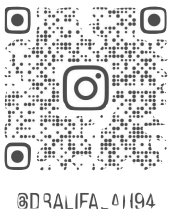


Sterilization in Operative Dentistry



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Is a term referring to any process that **eliminates (removes) or kills all forms of** microbial life, including the resistant bacterial spores using physical or chemical means.



DISINFECTION

procedure offering variable measures of control of infection, in this way we not going to kill all living microorganism may viruses kept survive.

Classification of sterilization

- ◆ PHYSICAL METHODS.
- ◆ CHEMICAL METHODS.
- ◆ COMBINATION OF BOTH.

PHYSICAL METHODS	CHEMICAL METHODS
Moist heat	Alcohols
Dry heat	Aldehydes
Filtration	Halogens
Radiation	Phenols



There are five accepted methods of sterilization:

1. Steam pressure sterilization (autoclave).
2. Chemical vapor pressure sterilization (chemi-clave).
3. Dry heat sterilization (dryclave).
4. Ethylene oxide sterilization.
5. Hydrogen Peroxide Plasma.



STEAM PRESSURE STERILIZATION (AUTOCLAVING)

- ◆ Sterilisation with steam under pressure
- ◆ (121° C) is a minimum of 15 minutes at 15 psi pressure
- ◆ 3 minutes if the temperature is raised to approximately (134°C) to give 20 psi of pressure.
- ◆ Instruments which are sterilized by using autoclave is Handpieces, orthodontic pliers, impression trays, and surgical instruments



CHEMICAL VAPOR PRESSURE STERILIZATION (CHEMICLAVING)

- ◆ Sterilization by chemical vapor under pressure is performed in a Chemiclave.
- ◆ Chemical vapor pressure sterilizers operate at (131° C) and 20 lbs pressure. About half an hour.
- ◆ ethylene oxide sterilizers, they must be used with a prescribed chemical by the manufacture
- ◆ Instruments which are sterilized by using chemiclave is:
 - Periodontal, restorative and endodontic instruments.
 - Carbon steel and other corrosion sensitive burs and pliers.



DRY HEAT STERILIZATION (hot air ovens)

- ◆ Dry heat sterilization is readily achieved at temperatures above **(160°C)** for **1-2 hours**.
- ◆ It is also fitted with a thermostat that maintains the chamber air at a chosen temperature.



Ethylene Oxide sterilization

- ◆ Ethylene oxide sterilization is the **best method for sterilizing complex, heat sensitive instruments and delicate materials like rotary handpiece**.
- ◆ Ethylene oxide is a **gas at a temperature below 100°C**.
- ◆ It is highly explosive and inflammable.
- ◆ It is used for the sterilization of **towels, metal and plastic instruments**.



LIQUID STERILANTS/HIGH-LEVEL DISINFECTANTS

- ◆ Sterilants used for high-level disinfection of items for reuse are **glutaraldehydes at 2% to 3%** concentrations; it kills bacterial spores in 6–10 hours.
- ◆ Glutaraldehydes are **irritating, sensitizing to skin and respiratory passages, and can be toxic as indicated in manufacturers' safety data sheets. Keep trays tightly covered in a well-vented area.**
- ◆ *Do not use 2% glutaraldehyde solutions to wipe counters or equipment (e.g., dental unit and chair).*

Infection control program in minimum dental office.

- 1. Sterilization of instruments**
- 2. Comprehensive medical history**
- 3. Hepatitis B vaccine to prevent any cross infection**
- 4. Antiseptic and mouthwash**
- 5. Disposal mask and gloves**
- 6. Protective eye glass**
- 7. Rubber dam**
- 8. Surface cleaning and cover tray**
- 9. Needle and sharp instrument safety**

DENTAL INSTRUMENTS CLASSIFICATION

based on risk of transmission and need of sterilization

- CRITICAL
- SEMI-CRITICAL
- NON-CRITICAL



Category	Definitions	Dental instrument or item	Type of sterilization
Critical	Penetrate soft tissue, contact bone, enters into or contacts the blood stream	Surgical instruments, periodontal scalers, scalpel blades, surgical dental burs.	HEAT STERILIZE between uses or use sterile single-use, DISPOSABLE devices
Semicritical	Contact mucous membranes or non intact skin, will not penetrate tissue, contact bone, enter into or contact blood stream.	Dental mouth mirror, amalgam condenser, reusable dental impression trays, dental handpieces	HEAT STERILIZE or HIGH-LEVEL DISINFECT
Noncritical	Contact intact skin	Radiograph head\cone, blood pressure cuff, facebow	Clean and disinfect using a LOW TO INTERMEDIATE LEVEL DISINFECTANT

Stages for instrument sterilization

- ◆ **Presoaking:** to dissolve or soften the debris
- ◆ **Cleaning:** ultrasonic cleaning is 9 times more effective than hand cleaning ,it provides fast and thorough cleaning without damage to instruments.
- ◆ **Packaging:**
- ◆ **Sterilization:** using steam autoclaving, dry-heat, or chemical vapor machines
- ◆ **Store Instruments in a Dry, Protected Area.**

CLEANING AND DISINFECTION OF DENTAL UNIT AND ENVIRONMENTAL SURFACES

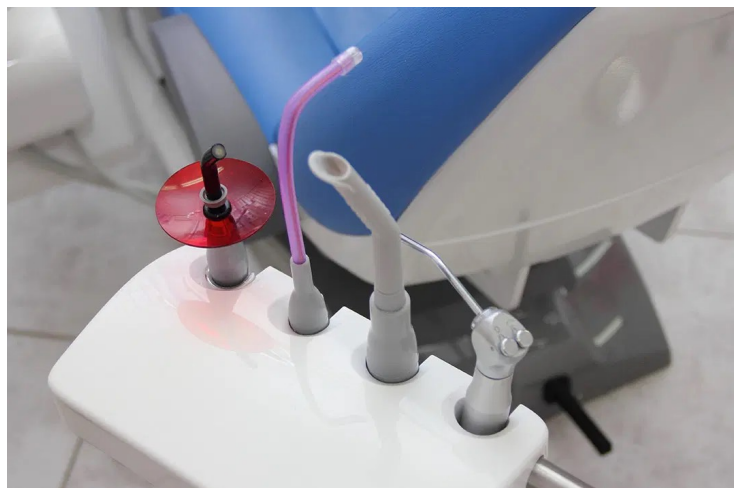
- ◆ Countertops and dental unit surfaces should be cleaned with disposable toweling
- ◆ Surfaces then should be disinfected with a suitable chemical germicide. Including: phenolics, iodophors, and chlorine-containing compounds.
- ◆ A fresh solution of sodium hypochlorite (household bleach) prepared daily Concentration (1/4 cup of bleach to 1 gallon of water) is effective on environmental surface, Caution should be exercised, since chlorine solutions are corrosive to metals, especially aluminum.

STERILISATION OF HANDPIECES

After each patient use:

- run any handpiece that is connected to the dental air/water system, to discharge water and/or air for at least 30 seconds
- Leave the bur in place while you clean the outside of the handpiece with detergent and warm water.
- Sterilize in an autoclave.
- If recommended by the manufacturer, lubricate the handpiece with pressurized oil until clean oil appears from handpiece.

handles or dental unit attachments of saliva ejectors, high-speed air evacuators, and air/water syringes. These components should be covered with disposable covers that are changed after each use



DISPOSAL OF WASTE MATERIALS

- Blood, suctioned fluids, or other liquid waste may be poured carefully into a drain connected to **a sanitary sewer system**.
- Disposable needles, scalpels, or other sharp items should be placed intact into **puncture-resistant containers** before disposal.
- Solid waste contaminated with blood or other body fluids should be **placed in sealed, sturdy impervious bags** to prevent leakage of the contained



Thank you